# CSC 360
# Operating Systems
# **Introduction**

Wenjun Yang

https://wenjun-y.github.io/csc360

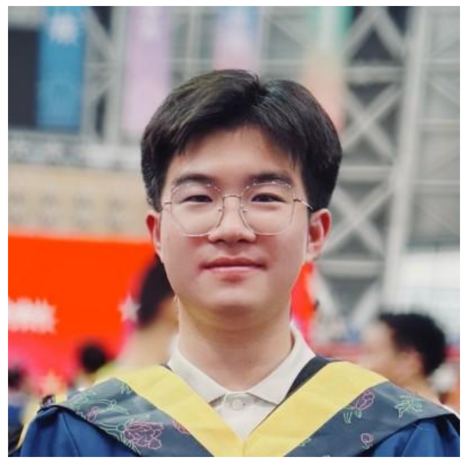
**Summer 2026**

# Instructor

- Dr. Wenjun Yang <wenjunyang@uvic.ca>
  - teacher-scholar & postdoc with CS
  - use UVic Teams first, as email not always reliable
    - to help, always include **[csc360]** in your email subject line
  - office hours: **W 12--1pm**
    - or by appointment
    - **on UVic Teams**
  - research area
    - Networked system for distributed AI
    - https://wenjun-y.github.io/

# CSC360's Mighty TAs

Mostafa Abdollahi



Quanwei Zhang

Ava Birtwistle

# Lab/tutorial instructors

- Mostafa, Quanwei, and Ava
  - their email on Brightspace
- Tutorials: start next week!
  - Please attend your registered section only!
    - tutorial lectures
      - **C**, libc, sockets, pthreads, ...
    - assignment help
      - spec go-through, common problems, ...
    - practice problems

# About the course

- *Introduction* to **Operating Systems**
  - (A01/2) **TWF 10:30--11:20am**, **ECS 125**
  - Bright: “Summer 2026 CSC 360 A01 A02 X”
    - assignments, gradebook, etc
    - **discussion channel hosted on UVic Teams**
  - prerequisites
    - Data structures (CSc **225** or 226)
    - Computer architecture (CSc **230** or CENG 255)
    - System programming (CSc/SENG **265**, CENG)

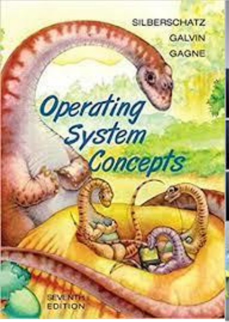
# Course evaluation

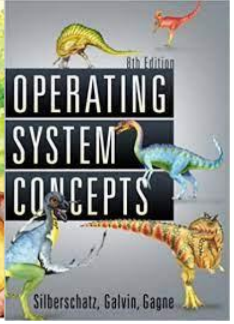
| **Assessment for CSC360:** | | |
|---|---|---|
| Assignment 1 | 5% | Due May 27 |
| Assignment 2 | 15% | Due June 3 |
| Assignment 3 | 15% | Due June 30 |
| Assignment 4 | 15% | Due July 31 |
| **Assignments Total** | **50%** | |
| Midterm 1 | 10% | June 5 |
| Midterm 2 | 10% | July 3 |
| Final Exam | 30% | TBD |
| **Exams Total** | **50%** | |

- Must-pass: A minimum score of 50% on the **final exam**

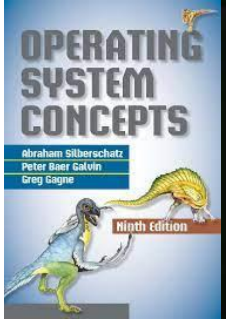
# Message from Undergrad Advisor

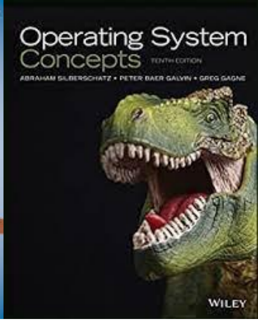
- Email: cscadvisor@uvic.ca
- Do not have the prerequisite course(s)?
  - need to obtain a waiver
  - otherwise, prerequisite drop after the first week
- Taking the course more than twice?
  - need to have a letter from the Chair and the Dean
  - otherwise, being dropped from the class
- Make sure you can receive email from Bright!
  - use UVic Teams first






# Course materials

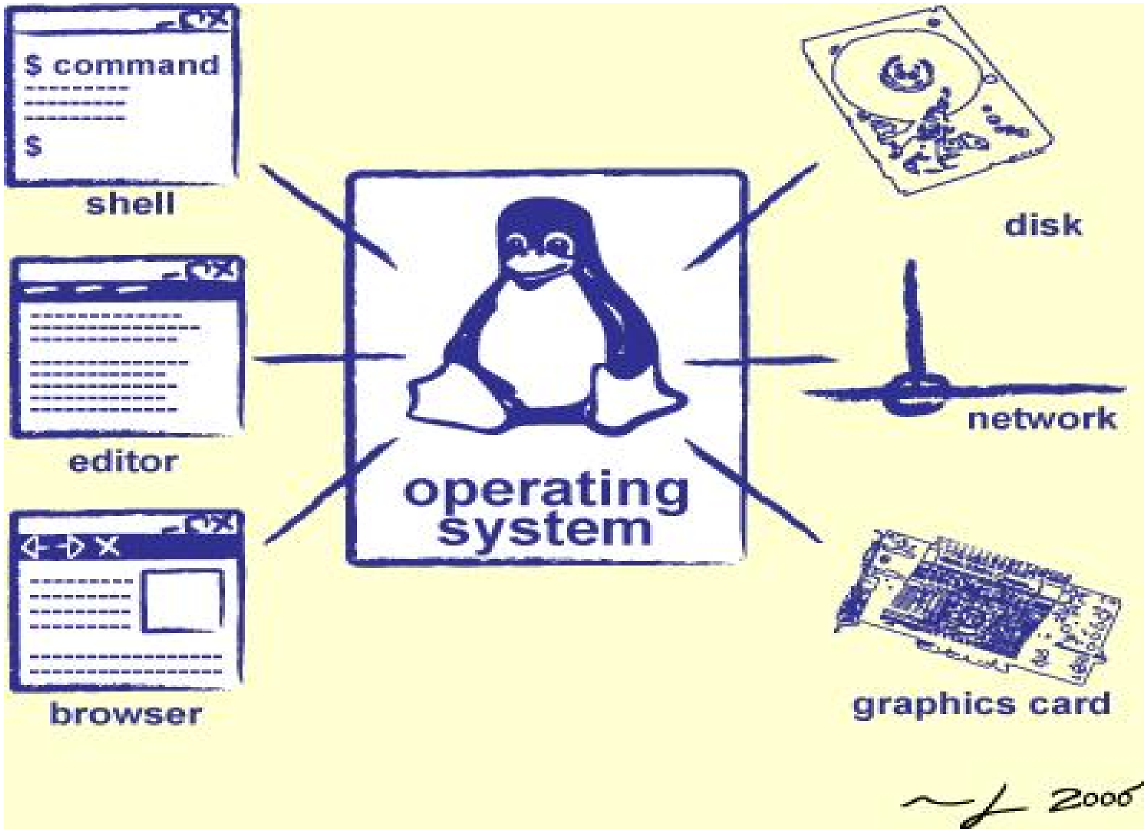




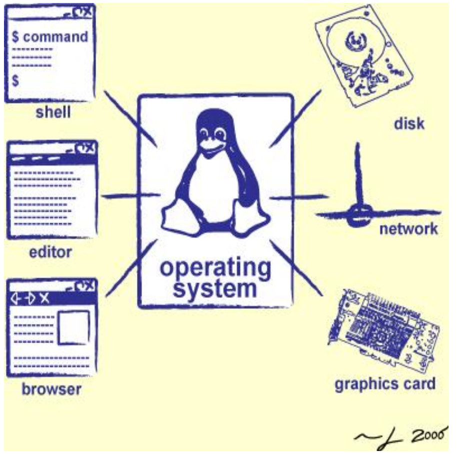

- Required textbook
  - Operating system concepts, 7th or **newer** editions
    - 6th and other editions: different chapter schedules
  - online resources

    http://codex.cs.yale.edu/avi/os-book/
    or **http://os-book.com/**
    - errata, slides, practice exercises and solutions
- Explore further
  - Google!

# Why take this course?



# Why take this course?

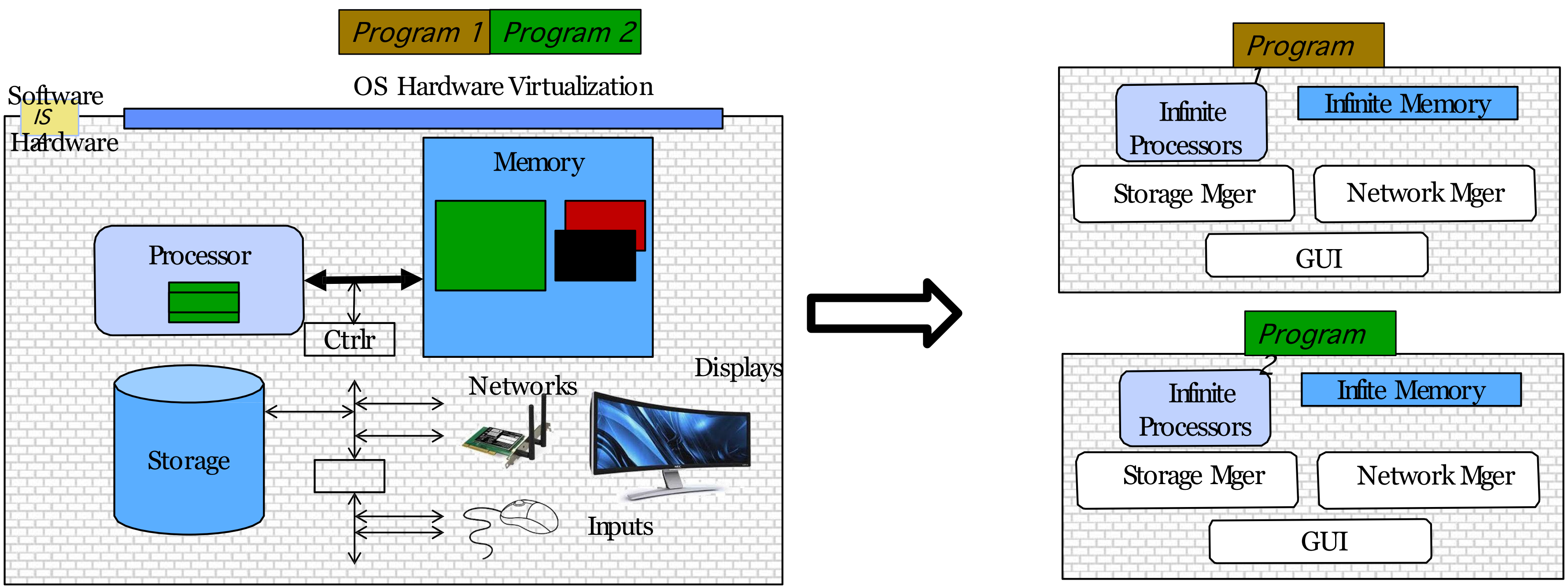



- The OS is everywhere
  - Every device, from your smartwatch, your smart light bulb, to your mobile phone and laptop runs an operating system
  - Every program you will ever write will run on an operating system
  - Its performance and execution behavior will depend on the operating system

# What is OS

Referee + illusionist + Glue

=> Easy to use virtual machine



# Three aspects for the Class

- Understanding OS principles
- System Programming
- Map Concepts to Real Code

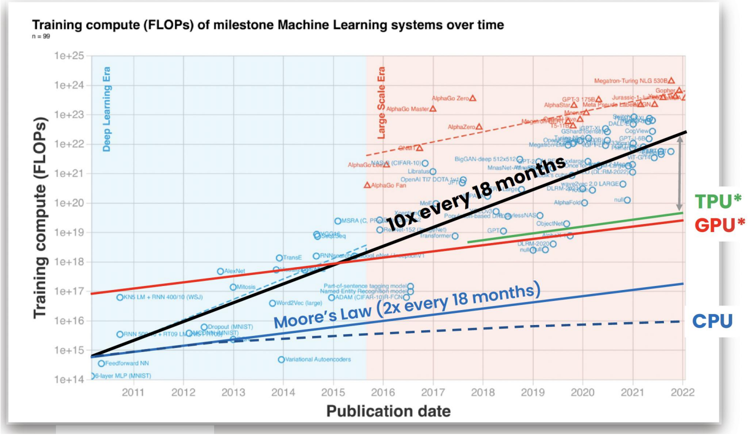
# Topic Breakdown (Course Objective)

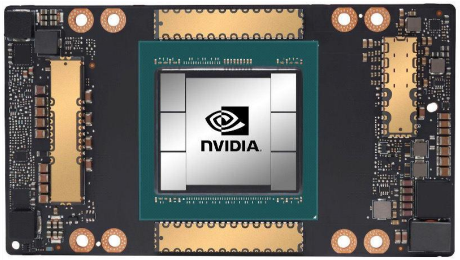
- virtualizing the CPU **(process)**: process, thread, scheduling, synch
- virtualizing memory (**memory)**: memory management, virtual memory
- **storage**: file systems, I/O systems

- Distributed system: communication, data processing, etc (not covered but falls into my research focus, happy to chat!)

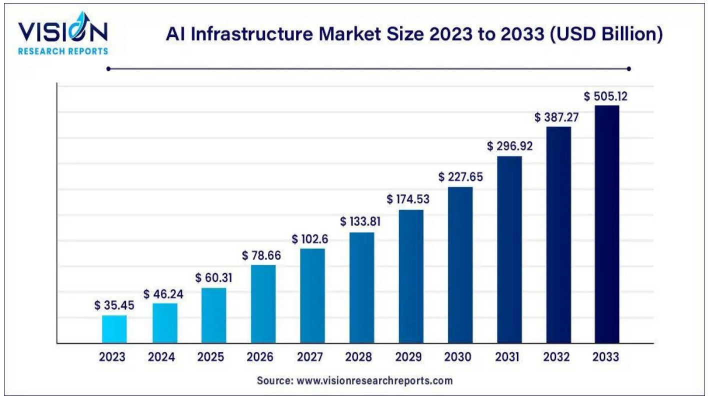
# One last note…

- Operating systems in the age of AI
- Everything gets reinvented
- This class more relevant than ever







# Lots of activity over past 15 years!

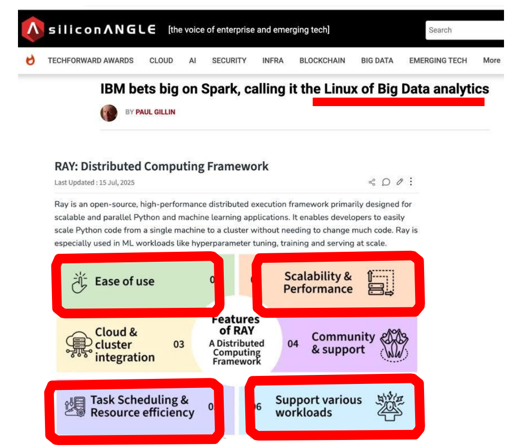
**How do you share a cluster across different distributed workloads ?**

**How do you share a cluster across different users, jobs, and queries?**

**How do you simplify the implementation and execution of distributed AI apps?**

**How do you do high performance LLM inference for many models across many GPUs supporting different workloads?**

**Mesosphere unveils Mesos-based datacenter OS plus $36m injection**




- **Memory management**
- **Scheduling, load balancing**
- **Performance isolation**
- **Hierarchical storage (GRAM, RAM, SSD)**
- **…**

# This lecture so far

- An introduction to the course
  - who, when, where, what, and why
  - course materials
  - course objectives
  - course topics

# Next lecture

- What is OS
- Interfaces to OS
  - CLI, GUI, system calls, API
  - read OSC7 Chapter 2 (or OSC6 Chapter 3)